Height Balanced Binary Tree

AlgoExpert · Medium · Binary Trees

The problem

- For every node, do left/right subtree heights differ by at most 1?
- Returns -1 from the helper as an "unbalanced" sentinel.

Intuition

- Compute heights bottom-up and fail fast if any node's subtrees differ by more than one.

Approach

- 1. Post-order returns height and balance
- 2. Compare child heights
- 3. Propagate any imbalance up

Java solution

```java
public class HeightBalancedBinaryTree {
    public static boolean heightBalancedBinaryTree(BinaryTree t) { return h(t) != -1; }

    private static int h(BinaryTree t) {
        if (t == null) return 0;
        int l = h(t.left), r = h(t.right);
        if (l == -1 || r == -1 || Math.abs(l - r) > 1) return -1;
        return 1 + Math.max(l, r);
    }
}
```

Complexity

- Time: O(n)
- Space: O(h)

Recap

- Compute heights bottom-up and fail fast if any node's subtrees differ by more than one.
- Complexity: O(n)